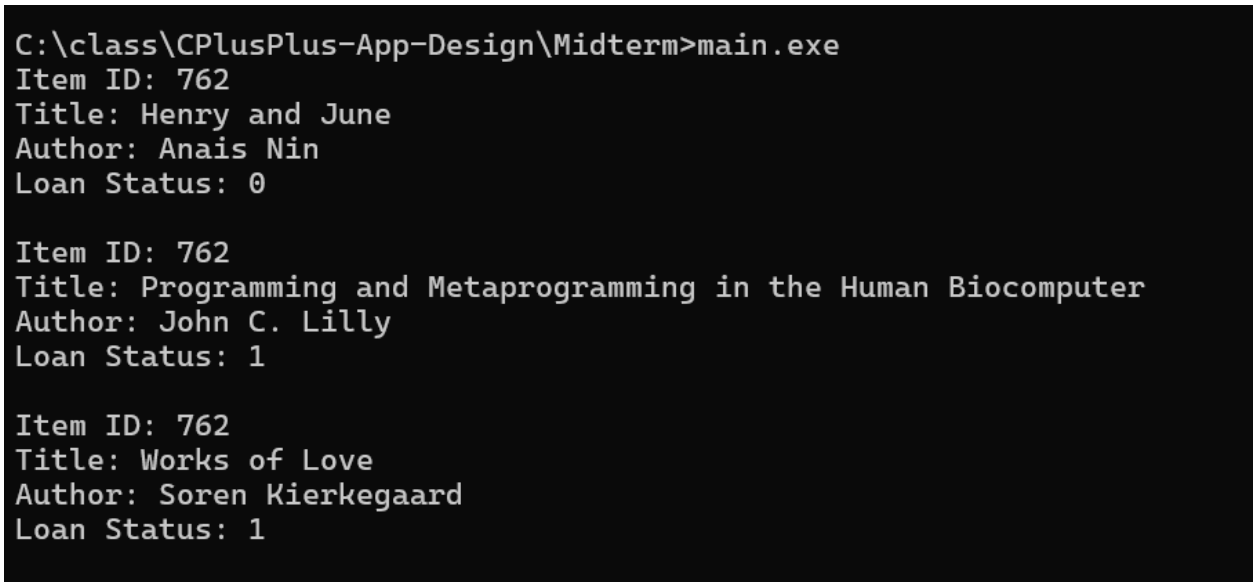
My midterm project simulates a very simple library database system, maintaining records of books (the only type of item the system currently accepts) with attributes for title, author, publisher, etc. Much like module 7's record keeping system, it uses two tiers of user types to divide privileges, with regular users only being able to display the current list of books, while the administrative user can add and delete books. If developed further, the Admin class would be outfitted with methods for checking out and returning books, by changing the enum value of the loanStatus attribute of any given book.

All memory is allocated automatically, so the size of data does not enter into the program logic explicitly anywhere, but it uses a variety of the usual types throughout (int, string, vector, etc). Vector is used as the collection type for the records object within main, to support passing by reference, rather than by value or copy, to ensure method calls are performing modifications on one and the same object. File input/output is minimally used, apart from records of books being added to a database file of csv type within the addBook method. Strings are used extensively throughout to contain data regarding the title, author, and so forth of books. To restate, two classes of user are used to divide privileges for acting on the data stored in the library's database.

Screenshots from a sample test run:



```
Enter the title of the book you would like to delete:
works of love
You would like to delete the book Works of Love by Soren Kierkegaard
 with library ID number 762?
Enter 'Y' to confirm, and any other key to delete another record:
y
Item ID: 762
Title: Henry and June
Author: Anais Nin
Loan Status: 0

Item ID: 762
Title: Programming and Metaprogramming in the Human Biocomputer
Author: John C. Lilly
Loan Status: 1
```